

Linear Algebra

Summer 2026

Homework 4

Deadline: 22/06/2026

Problem 1 (2 POINTS)

Show that the transformation T defined by $T(x_1, x_2, x_3) = (x_1 - 2x_2, x_1 - 3x_2, x_3 - 5, 0)$ is not linear.

Problem 2 (3.5 POINTS)

$$A = \begin{bmatrix} 3 & 4 & -7 & 0 \\ 5 & -8 & 7 & 4 \\ 6 & -8 & 6 & 4 \\ 9 & -7 & -2 & 0 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 4 \\ -4 \\ -4 \\ -7 \end{bmatrix}.$$

[P] Is b in the range of the transformation $x \mapsto Ax$?

If so, find an x whose image under T is b .

Do not forget to include your Python commands.

Problem 3 (5.5 POINTS)

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation with $T(x_1, x_2) = (x_1 - 2x_2, -x_1 + 3x_2, 3x_1 - 2x_2)$.

- (1 POINT) Find the standard matrix of T .
- (2 POINTS) Find x such that $T(x) = (-1, 4, 9)$.
- (2.5 POINTS) Is T onto? Is it one-to-one? Explain.